

Strategy Typology and Cost Structure: A Textual Analysis Approach

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Abstract

This study investigates the impact of firms' strategic positioning on cost structure. Using textual analysis based on 10-K filings, we capture business strategy along three dimensions: product leadership, customer intimacy, and operational excellence. Firms pursuing the product leadership strategy emphasize innovation and confront high congestion risk caused by rapid growth. We find that these firms display a more rigid cost structure by incurring higher strategic fixed investments in R&D and choosing higher committed capacity to reduce congestion risk. We also document that firms following the other two strategies on average exhibit a less rigid cost structure. This is possibly because customer-intimate firms tend to invest in specialized resources in small chunks and take on-demand inputs to tailor offerings, while operational-excellence firms may adopt real options and flexible models to achieve production efficiency and mitigate default risk. Our results are robust to using a difference-in-difference approach based on a quasinalatural shock caused by the Inevitable Disclosure Doctrine adoption and a change analysis based on firms' internal strategic shifts. Our findings suggest that organizational strategy has a significant impact on firms' resource commitments and capacity choices.

Keywords

Business strategy, textual analysis, cost structure, cost rigidity

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1 Introduction

Cost structure refers to the relative proportion of fixed and variable costs and has significant implications for firms' operational flexibility and financial performance (Aboody et al., 2018; Banker and Chen, 2006; Bhojraj et al., 2021). As such, it is important to understand the factors contributing to firms' cost structure decisions, which reflect their choices on resource commitments and capacity levels. A growing body of accounting literature has identified various determinants of cost structure, such as demand conditions (Banker et al., 2014), supply chain relationship (Chang et al., 2021), managerial incentives (Aboody et al., 2018), and managerial ability (Banker et al. 2024). In addition to the above factors, business strategy may also affect cost structure because it guides firms' daily operations and resource optimization (Hambrick, 1983a; Schendel and Hatten, 1972). However, the effect of business strategy on cost structure has been understudied. To fill this void, we use textual analysis to measure business strategy and explore its essential role in firms' resource deployment and cost management decisions.

In this paper, we consider three organizational strategies proposed by Miller (1986) and Treacy and Wiersema (1995): product leadership, customer intimacy, and operational excellence.¹ Building on theories about the three strategies, we explore their possible effects on cost structure via strategic investments and demand conditions. First, firms excelling in product leadership continuously explore new opportunities and develop innovative products with intensive R&D activities (Hambrick, 1983a; Nair and Filer, 2003; Porter, 1985; Spanos et al., 2004). R&D projects require long-term commitments in human capital, fixed assets, and intangible assets, which lead to fixed costs and hence a more

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rigid cost structure (Hall et al., 1986; Brown et al., 2009; Hall and Lerner, 2010). Additionally, product leaders' quick response to changing market conditions allows them to grow in spurts (Miles and Snow, 1978, 2003). Facing consequential high congestion risk, they tend to choose a higher capacity level to relieve congestion, implying a more rigid cost structure (Banker et al., 2014). However, it is possible that product leaders will display a less rigid cost structure through focusing on core technology development and outsourcing the manufacturing of nonessential components. As such, they could maintain flexibility in resource management and mitigate downside risk driven by unfavorable R&D outcomes. Therefore, the association between product leadership and cost rigidity is an empirical question.

Second, firms leaning toward customer intimacy provide tailored products or services to build customer loyalty (Treacy and Wiersema, 1995). To support durable customer relationships, they may invest in resources that are dedicated to a particular relationship, contributing to a more rigid cost structure (Irvine et al., 2016). Alternatively, to fulfill individual customer requests, they may only use variable or on-demand inputs with lower upfront commitments, which enable them to manage downside risk associated with customer defection and exhibit a less rigid cost structure. The above two conflicting predictions suggest that the association between customer intimacy and cost rigidity can be positive or negative.

Third, firms that have embraced an operational-excellence strategy focus on efficiency and compete on the basis of price (Hambrick, 1983a; Porter, 1985). They may invest heavily in physical assets to build economies of scale, thereby spreading fixed costs over a high volume of output and lowering unit cost (Porter, 1985). Otherwise, they could adopt real options (i.e., operating leases or contract labor) to maintain flexibility in capacity building or a just-in-time model to achieve production and distribution efficiency (Kallapur and Eldenburg, 2005). These approaches allow them to alleviate downside risk due to substitutes provided by competitors and imply a less rigid cost structure. Taken together, operation-excellence firms may exhibit a higher or lower level of cost rigidity.

We use textual analysis to construct a multidimensional strategy measure for a broad cross-section of firms and industries (Banker and Ma, 2018). We extract Item 1-Business in 10-K filings for all public firms from 1993 to 2019 and develop a keyword list to capture firms' self-description of strategic priorities and corresponding functional activities. We count the keyword frequency for each 10-K report and conduct an exploratory factor analysis to reduce the dimensionality of keywords. We retain three output factors with eigenvalues greater than one and then compute weighted factor scores. After reviewing keywords that load high on each factor, we find that they align with the strategy classification introduced by Miller (1986) and Treacy and Wiersema (1995). We follow Treacy and Wiersema (1995) to label the three output factors as product leadership, customer intimacy, and operational excellence, respectively.

To estimate a firm's cost structure in a given year, we regress the log change in total operating costs on the log change in sales revenue, where a lower slope coefficient suggests a higher level of cost rigidity (Aboody et al., 2018; Banker et al., 2014; Kallapur and Eldenburg, 2005).² We interact our strategy measures with the log change in sales and investigate the association between business strategy and cost structure. First, we find that, on average, the product leadership strategy is positively associated with cost rigidity, suggesting product leaders' greater emphasis on R&D activities and upside congestion risk. In contrast, firms following the other two strategies on average exhibit a lower level of cost rigidity. It could be because customer-intimate firms are inclined to rely on variable and on-demand inputs to tailor offerings, while operational-excellence firms are likely to utilize real options and flexible models to build up capacity, control costs, and manage downside risk. Second, we exploit the adoption of the Inevitable Disclosure Doctrine (IDD) as an exogenous shock to firms' orientation toward intended strategies. We confirm that firms experience increases in strategy scores around IDD adoption and find consistent results using a difference-in-difference research design. Third, we conduct a change analysis based on firms' internal strategic shifts and find that an increase in firms' orientation toward product leadership (customer intimacy or operational excellence) will give rise to an increase (a decrease) in cost rigidity. Fourth, we document that business strategy affects cost rigidity through the channels of strategic investments and demand conditions. Fifth, we find that the association between strategy and cost structure is intensified by market share and demand uncertainty. Lastly, we observe that extremely large operational-excellence firms may operate with a higher production level to reach economies of scale, resulting in a more rigid cost structure.

Our study contributes to several strands of literature. First, our study adds to the operations management and accounting literature on cost structure. The existing literature has predominantly focused on external or managerial factors that drive firms' cost structure decisions (Aboody et al., 2018; Banker et al., 2014; Chang et al., 2021; Holzhacker et al., 2015a, b). Our study extends this line of research by focusing on a firm's internal strategic choice, which directs managers to make operational decisions and organize available resources. It highlights that strategy could be a fundamental determinant of cost structure, thereby providing useful guidance for managerial capacity choices and capital market risk assessment.

Second, we contribute to the operations management and accounting literature on the real effects of business strategy on corporate decisions. This stream of literature has explored the critical role of strategy in enhancing operational efficiency (Dess and Davis, 1984; Hambrick, 1983a; Zott and Amit, 2008), facilitating supply chain collaboration (Chang et al., 2022), and designing CEO compensation (Ittner et al., 1997). However, the empirical evidence on whether and how the generic strategy cohesively bonds a business plan with resources and capacity is scarce. We extend the research on

business strategy to resource arrangements, capacity commitments, and corresponding cost behavior.

Third, this study improves our understanding of the implementation of business strategy. Previous literature documents that an organization's strategy involves a set of plans and actions to accomplish its goals (Hambrick, 1983a; Porter, 1985). However, there is limited empirical evidence on how firms execute business strategy at the department level, partially because it is difficult to measure strategy for a broad cross-section of firms.³ In this paper, we use a textual measure of strategy based on corporate disclosures, which is widely available and comparable across firms and industries. Employing a large sample of 62,116 observations, our results suggest that the management will choose the optimal capacity level in the production department and organize resources in multiple functional departments (i.e., R&D and human resources) to implement strategy and accomplish common goals. Therefore, this study sheds new insights on how strategy drives the coordination and collaboration among departments in resource arrangements.

The remainder of this paper is structured as follows. Section 2 reviews literature and develops hypotheses. Section 3 describes data and empirical specifications. Section 4 presents main results and Section 5 discusses additional analyses. Section 6 concludes.

2 Hypothesis Development

To achieve the desired competitive position in an industry, firms should choose a generic strategy and execute it over the long term (Porter, 1980). The implementation of strategy requires managers to deploy valuable resources (i.e., employees and fixed assets) in multiple functional departments to support value-creating activities (Schendel and Hatten, 1972), which may subsequently influence the mix of fixed and variable costs. Therefore, business strategy can be a fundamental determinant of cost structure. However, prior research on how strategy affects cost structure is limited. We seek to fill this void.

Firms' competitive strategies could influence the level of cost rigidity via strategic investments and demand conditions. We expect that firms' orientation toward product leadership may be positively associated with cost rigidity due to the following two reasons. First, product leaders strive to provide innovative products and services to stay ahead of the competition. With a strong emphasis on innovation, they tend to undertake substantial investments in R&D activities (Hambrick, 1983a; Ittner et al., 1997; Treacy and Wiersema, 1995). R&D projects often rely on large and ongoing investments in committed resources, including human capital (i.e., high-skilled talents) and physical capital (i.e., expensive equipment) (Banker et al., 2019; Hall et al., 1986; Hall and Lerner, 2010).⁴ These dedicated resources are firm-specific and not readily transferable or replaceable. Therefore, they are costly to adjust and should be committed in advance, leading to a high level of fixed costs (Brown et al., 2009; Peters and Taylor, 2017).

Product leaders may also choose to incur fixed R&D expenditures that serve as a commitment device, thereby conveying to the market that firms will stand by their innovation and maintain the existing business course.⁵ As a result, product leaders may operate with a higher proportion of fixed costs, indicating a more rigid cost structure.

Second, product leaders tend to experience large outcome uncertainty due to the highly risky nature of R&D investments, leading to fluctuating demand conditions (Hall and Lerner, 2010; Miles and Snow, 1978, 2003). At the same time, continuous development in technologies enables product leaders to respond quickly to market changes, exploit new product features, and generate demand surges (Miles and Snow, 1978, 2003; Treacy and Wiersema, 1995). High demand realizations coupled with large demand uncertainty will engender severe congestion if firms are operating with a limited capacity of fixed inputs (Banker et al., 2014). Anticipating rapid demand growth and potential congestion issues, they may commit to a higher capacity level in advance and operate with a more rigid cost structure (Banker et al., 2014). Therefore, considering the effects of strategic fixed investment and upside congestion risk, we state the following hypothesis.

H1a: Firms' orientation toward a product leadership strategy is positively associated with cost rigidity.

Nevertheless, firms' strategic positioning as a product leader may be associated with a less rigid cost structure. These firms offer creative products with cutting-edge features and short life cycles. Therefore, they are likely to maintain flexibility in resource deployment to accommodate emerging technological trends and quick product changes (Chang et al., 2022). Product leaders may focus on key technologies and patents while outsourcing the manufacturing of nonessential and standard components. For instance, in 2002, Hewlett-Packard Company announced that it would outsource its PC manufacturing facilities worldwide while keeping technology development in-house, thereby strengthening strategic core competencies in designing and developing new products. This decision would allow it to "take advantage of cost benefits associated with using non-dedicated factories" and "leverage cost structure".⁶ In addition, product leaders may consider outsourcing noncore business services (i.e., machine maintenance, engineering, accounting, and information systems) to convert fixed costs, such as capital expenditures and fixed employee salaries, into variable costs (Abraham and Taylor, 1996; Kallapur and Eldenburg, 2005). Adopting these flexible operating models and options, they could not only prioritize product design and development but also mitigate downside risk if they confront unfavorable R&D outcomes and consequential business downturns. Hence, product leaders may choose the above-mentioned models with corresponding investment deployment, engendering a less rigid cost structure. We state the following hypothesis associated with flexible resource management and downside default risk.

H1b: Firms' orientation toward a product leadership strategy is negatively associated with cost rigidity.

Next, we explore the association between the customer intimacy strategy and cost rigidity. Customer-intimate firms generate benefits through tailoring offerings and building customer loyalty in the long term. To initiate and promote a stable customer relationship, customer-intimate firms may undertake large relationship-specific investments by incurring high fixed costs. For example, they may purchase specialized equipment to tailor products, train employees to service the products, or choose specific production sites to satisfy customer demands (Titman and Wessels, 1988). These investments are dedicated to a particular relationship and are less transferable than general investments, thereby giving rise to fixed costs (Chang et al., 2021; Irvine et al., 2016). At the same time, customer-intimate firms tend to invest heavily in marketing and advertising activities to identify customer needs and increase brand recognition (Miller, 1986; Treacy and Wiersema, 1995). Investments in these areas are considered to be fixed because they indicate a budget allocation over a particular period and remain unchanged till the next planning cycle. In addition, customer-intimate firms may be concerned about upside congestion risk associated with increasing sales from loyal customers, which incentivizes them to increase committed capacity levels to relieve congestion. As a result, customer-intimate firms may operate with a more rigid cost structure under the influence of strategic fixed investment and upside congestion risk. Accordingly, we state the following hypothesis.

H2a: Firms' orientation toward a customer intimacy strategy is positively associated with cost rigidity.

Alternatively, the customer intimacy strategy requires that companies "continually tailor and shape products and services to fit an increasingly fine definition of the customer" (Treacy and Wiersema, 1993). That is, they aim at meeting each customer's needs separately instead of serving a large group of general customers that share the same needs. Therefore, they may operate with various modes of producing and delivering goods or services (Treacy and Wiersema, 1993). For instance, Kraft USA, an example of a customer-intimate firm, decentralizes its operations and tailors its merchandise and human assets (i.e., educated sales force) to satisfy the needs of a particular store's customers (Chang et al., 2022; Treacy and Wiersema, 1993). As such, their investments are in small chunks, which can be adjusted in a nimble manner, rather than in lumpy and large assets for general purposes. Moreover, they may reduce commitments at the early stage for individual customers but only use on-demand inputs to provide highly customized products in small quantities. Specifically, they could utilize advanced information systems and workforce responsiveness to collect reliable customer information, analyze customer preferences, and address customer inquiries (Treacy and Wiersema, 1993; Chang et al., 2022). With real-time data analytics, they could minimize initial fixed investments and develop programs to fit special requirements immediately and precisely. For example, a build-to-order model is widely employed

to offer mass-customized products upon receiving orders, thereby allowing customer-intimate firms to achieve individual customization flexibility and capacity utilization efficiency (Chang et al., 2022). These operating models and resource management also enable them to address downside risk associated with customer defection, because the corresponding investments are less costly to adjust. Altogether, firms excelling in customer intimacy may display a less rigid cost structure. We state the hypothesis on flexible resource management and downside default risk as follows:

H2b: Firms' orientation toward a customer intimacy strategy is negatively associated with cost rigidity.

Operational-excellence firms compete by providing undifferentiated products and services with low prices (Miles and Snow, 1978, 2003; Porter, 1985; Treacy and Wiersema, 1995). To achieve cost minimization, these firms could take advantage of economies of scale with substantial investments in physical resources, such as machinery and equipment, and spread fixed costs over large numbers of units (Porter, 1985). As a result, they could smooth out frictions in the conversion process and reduce transaction costs in market exchange. At the same time, they may pursue technological efficiency by offering high volumes of standardized products, resulting in a high degree of mechanization and routinization (Bentley et al., 2013). This may further create constraints that force them to rely on physical resources for subsequent maintenance and continuous improvement, which are considered to be fixed in nature (Bentley et al., 2013). In addition, these firms may increase capacity levels to address potential congestion risk associated with increasing sales from selling low-cost products and services. Therefore, firms following the operational excellence strategy may primarily spend on physical capital and exhibit a more rigid cost structure. We state the following hypothesis related to strategic fixed investment and upside congestion risk:

H3a: Firms' orientation toward an operational excellence strategy is positively associated with cost rigidity.

However, operational-excellence firms may prefer a less rigid cost structure. It is crucial for operational-excellence firms to maintain flexibility in resource planning and capacity building. As a result, they could adjust costs in response to demand changes timely and control costs effectively. To accomplish this goal, they could utilize some real options to minimize the upfront fixed investments and build up manufacturing capacity flexibly (Kallapur and Eldenburg, 2005). For example, firms may rent plants and machinery through operating leases and return these facilities if demand realizations fall (Kallapur and Eldenburg, 2005). They may also increase the proportion of short-term contract labor, which could be scaled down conveniently through pausing new hires and discontinuing existing contracts (Kallapur and Eldenburg, 2005). These options can transform fixed costs into variable costs and thus "variabilize" cost structure. Furthermore, with a primary focus on production efficiency, firms (i.e., Toyota) may adopt a just-in-time model to produce the exact amount needed at the exact

Table 1. Sample selection and sample distribution.

<i>Panel A: Sample selection</i>		Number of Observations	
Unique observations with Item 1 of the 10-K report to construct textual variable from 1993 to 2019		144,285	
Drop observations unidentifiable from Compustat		(44,282)	
Drop observations in utilities (SIC 4900-4999) and financial (SIC 6000-6099) industries		(28,022)	
Drop observations with missing data on control variables		(9865)	
Final Sample		62,116	
<i>Panel B: Sample distribution by industry</i>		Number of	
Industry group name		Observations	Frequency
Consumer nondurables—food, tobacco, textiles, apparel, leather, toys		3096	4.98%
Consumer durables—cars, TV's, furniture, household appliances		1925	3.10%
Manufacturing—machinery, trucks, planes, off furn, paper, Com printing		7797	12.55%
Energy—oil, gas, and coal extraction and products		3838	6.18%
Chemicals and allied products		2046	3.29%
Business equipment—computers, software, and electronic equipment		15,423	24.83%
Telecom—telephone and television transmission		2429	3.91%
Shops—wholesale, retail, and some services (laundries, repair shops)		5171	8.32%
Healthcare, medical equipment, and drugs		10,474	16.86%
Other—mines, Constr, BldMt, Trans, hotels, Bus Serv, entertainment		9917	15.97%
Total		62,116	100.00%

Notes: This table reports sample selection procedures in Panel A and sample distribution by industry based on Fama and French (1997) in Panel B. The sample spreads from 1993 to 2019.

time with successful supply chain coordination. As a result, they could manufacture in smaller batches at a lower capacity level, which saves fixed costs and predicts a lower level of rigidity. Importantly, as operational-excellence firms sell products and services with readily available substitutes, they may care about downside risk in the presence of competitors (Bentley et al., 2013; Miles and Snow, 1978, 2003). These flexible operating models and resource management could allow them to alleviate downside risk. Taken together, due to the preference for flexibility and heightened concern for downside risk, operational-excellence firms may opt for arrangements that result in a less rigid cost structure. Therefore, we state the following hypothesis.

H3b: Firms' orientation toward an operational excellence strategy is negatively associated with cost rigidity.

3 Data and Empirical Specification

3.1 Data

Following Banker and Ma (2018), we identify firms' generic strategies based on textual analysis of their 10-K filings from 1993 to 2019.⁷ The 10-K filings are regular, formal disclosure filed under Securities and Exchange Commission regulations (SEC 2011) and contain detailed and private information on firms' business, risks, and operations. As a result, the language and forms of 10-K filings are comparable across a large cross-section of firms, allowing us to conduct textual analysis to parse out qualitative information (Feldman et al., 2010; Kothari et al., 2009; Loughran and McDonald, 2016). We

concentrate on Item 1-Business of Form 10-K, because this section requires the management to provide a standardized and thorough description of firms' products, services, and strategic positioning (Hoberg and Phillips, 2016). This description is mandated by Regulation S-K to provide information about firms' current and future business plans, which can be used to capture their strategy (SEC 2016).

Table 1, panel A presents the sample selection process. Our sample starts with all 10-K filings from the SEC's EDGAR database for fiscal years 1993–2019. After extracting Item 1 from each 10-K report and excluding Item 1 sections that are shorter than 1000 words, we obtain an initial sample of 144,285 firm-year observations to construct strategy measures.⁸ We then link them to Compustat firms based on Central Index Key and filing dates. Firms in utilities and financial industries (SIC 4900-4999 and 6000-6799) are removed due to their unique characteristics. We further drop observations with missing financial data on variables required to estimate our empirical models, resulting in a final sample of 62,116 firm-year observations. Table 1, panel B outlines the sample distribution by industry based on Fama and French (1997). In our sample, business equipment (24.83%), healthcare (16.86%), manufacturing (12.55%), and others (15.97%) are the four largest industries that contain more than 10% of total observations.

3.2 Measuring Strategy Using Textual Analysis

Appendix A provides a detailed description of variable definitions. Consistent with prior literature in linguistics and

Table 2. Summary statistics and correlations.

Panel A: Summary statistics						
Variable	N	Mean	25%	Median	75%	St Dev.
$\Delta \ln \text{SALES}$	62,116	0.1103	−0.0365	0.0796	0.2293	0.4558
$\Delta \ln \text{XOPR}$	62,116	0.1023	−0.0254	0.0781	0.2132	0.3377
<i>PRODUCT</i>	62,116	0.0100	−0.7139	−0.1521	0.5239	0.9026
<i>CUSTOMER</i>	62,116	0.0180	−0.6223	−0.2550	0.3468	0.9333
<i>OPERATION</i>	62,116	0.0198	−0.6290	−0.2685	0.4071	0.9471
<i>UNCERT</i>	62,116	0.3977	0.1490	0.2460	0.4439	0.4254
<i>AI</i>	62,116	3.5960	0.7292	1.1394	2.0425	12.4294
<i>EI</i>	62,116	9.5188	2.6346	4.6338	7.8777	23.4049
<i>AGE</i>	62,116	2.6009	2.0794	2.5649	3.1781	0.7827
<i>MVE</i>	62,116	5.5238	3.9194	5.5568	7.1121	2.3044
<i>GROWTH</i>	62,116	0.2705	−0.0359	0.0829	0.2577	1.0087
<i>LEV</i>	62,116	0.1994	0.0008	0.1215	0.3134	0.2451

Panel B: Pearson and Spearman correlations between variables												
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
(1) $\Delta \ln \text{SALES}$		0.77	0.04	0.02	−0.02	0.04	0.04	−0.03	−0.20	0.16	1.00	−0.01
(2) $\Delta \ln \text{XOPR}$	0.65		0.03	0.01	−0.01	0.04	0.09	0.01	−0.22	0.17	0.77	−0.01
(3) <i>PRODUCT</i>	0.02	0.01		0.08	0.04	0.09	0.05	0.18	−0.13	−0.03	0.04	−0.26
(4) <i>CUSTOMER</i>	0.02	0.02	0.01		−0.00	−0.11	−0.13	−0.01	−0.08	0.03	0.02	−0.07
(5) <i>OPERATION</i>	−0.01	−0.01	0.03	0.00		−0.12	−0.16	−0.00	−0.00	0.03	−0.02	0.08
(6) <i>UNCERT</i>	0.07	0.05	0.07	−0.09	−0.13		0.32	0.04	−0.25	−0.32	0.04	−0.15
(7) <i>AI</i>	−0.13	0.03	0.04	−0.07	−0.09	0.39		0.10	−0.13	0.17	0.04	0.06
(8) <i>EI</i>	−0.15	0.01	0.07	−0.04	−0.05	0.34	0.74		−0.12	−0.29	−0.03	−0.10
(9) <i>AGE</i>	−0.18	−0.21	−0.12	−0.07	0.02	−0.22	−0.11	−0.11		0.23	−0.20	0.13
(10) <i>MVE</i>	0.12	0.14	−0.03	0.04	0.03	−0.26	−0.05	−0.15	0.24		0.16	0.21
(11) <i>GROWTH</i>	0.84	0.50	0.03	−0.00	−0.04	0.30	0.05	0.02	−0.19	0.01		−0.01
(12) <i>LEV</i>	−0.01	−0.02	−0.19	−0.02	0.03	−0.03	−0.03	−0.03	0.06	0.08	−0.02	

Notes: This table reports summary statistics of key variables for the full sample in Panel A and Pearson (lower left) and Spearman (upper right) correlations in Panel B. The full sample contains 62,116 observations from 1993 to 2019. Variable definitions are presented in Appendix A. All continuous variables are winsorized at the 1% and 99% levels to reduce the influence of outliers. Bold figures indicate significant levels of less than 5%. AI = asset intensity; EI = employee intensity; MVE = market capitalization; LEV = leverage.

strategy, we use a word-count approach to measure strategy, because word frequency suggests the intensity of focus and concern (Abrahamson and Hambrick, 1997; Huff, 1990; Kaplan, 2008; Weber, 1990). To create a word list for frequency counts, we consider words and phrases associated with business strategy from two perspectives: the conceptual description and relevant value-creating activities. To ensure that keywords are selected with a solid theoretical foundation, we develop the list based on Porter (1985) and other studies on organizational strategy. Therefore, we build on prior research to evaluate the usefulness of each keyword and establish the content validity. Appendix B provides a keywords list of eight groups and citations for each keyword selected in our list.⁹

We follow three steps to create our strategy measures (Banker and Ma, 2018). First, for each firm-year observation, we extract Item 1-Business, count the number of keywords, categorize the keyword counts into eight groups as shown in Appendix B, and scale the keyword counts by the total number of words in Item 1. Next, we conduct an exploratory factor analysis of keyword counts without specifying the number of

output factors. This analysis yields three latent factors with eigenvalues greater than one, which together explain around half of the sample variance. Appendix C reports the results of our factor analysis with varimax rotation.¹⁰ We manually check the keywords that load high on each factor and find that the three output factors are consistent with the strategy typology in Miller (1986) and Treacy and Wiersema (1995), further providing a validation of our keyword selection. Accordingly, we label the three factors as *PRODUCT* for the product leadership strategy, *CUSTOMER* for the customer intimacy strategy, and *OPERATION* for the operational efficiency strategy.¹¹ Lastly, we use factor loadings to compute weighted factor scores for the three strategies, whereas a higher score indicates that the firm leans more toward the corresponding strategy.¹² Table 2 presents summary statistics for the three strategy measures.^{13,14}

3.3 Empirical Model

Following prior studies, we capture cost structure as the percent change in costs for a one percent change in sales using a time-series regression model (Aboody et al., 2018;

Banker et al., 2014; Kallapur and Eldenburg, 2005). Specifically, we employ the following model:

$$\begin{aligned} \Delta \ln XOPR_{i,t} = & \beta_0 + \beta_1 \Delta \ln SALES_{i,t} \\ & + \beta_2 PRODUCT_{i,t} * \Delta \ln SALES_{i,t} \\ & + \beta_3 CUSTOMER_{i,t} * \Delta \ln SALES_{i,t} \\ & + \beta_4 OPERATION_{i,t} * \Delta \ln SALES_{i,t} \\ & + \sum_{m=5}^{11} \beta_m Control_{i,t} * \Delta \ln SALES_{i,t} \\ & + \beta_{12} PRODUCT_{i,t} + \beta_{13} CUSTOMER_{i,t} \\ & + \beta_{14} OPERATION_{i,t} + \sum_{n=15}^{21} \beta_n Control_{i,t} + \epsilon_{i,t} \end{aligned} \quad (1)$$

where $\Delta \ln XOPR_{i,t}$ is the log-changes in total operating costs for firm i from year $t-1$ to t ; $\Delta \ln SALES_{i,t}$ is the log-changes in sales for firm i from year $t-1$ to t ; $PRODUCT_{i,t}$, $CUSTOMER_{i,t}$, and $OPERATION_{i,t}$ are our textual measures of three strategies. To investigate the effects of three strategies on cost rigidity, we focus on the coefficients on interactions between strategy scores and $\Delta \ln SALES_{i,t}$.

We include common controls for cost structure. First, as in Banker et al. (2014), we control for demand uncertainty calculated as the standard deviation of $\Delta \ln SALES$ for all valid observations of firm i ($UNCERT_i$). Second, consistent with Anderson et al. (2003), we control for adjustment costs of underlying resources measured by asset intensity ($AI_{i,t}$) and employee intensity ($EI_{i,t}$). Third, following Banker et al. (2014) and Irvine et al. (2016), we include firm characteristics that may influence cost structure in the long run, such as firm age ($AGE_{i,t}$), market capitalization ($MVE_{i,t}$), sales growth ($SGROWTH_{i,t}$), and leverage ($LEV_{i,t}$). Finally, to control for omitted year effects and industry characteristics, we include year and industry fixed effects at the two-digit SIC level. Standard errors are clustered at the firm level. We winsorize all continuous variables at the top and bottom one percent levels to reduce the influence of outliers.

3.4 Summary Statistics

Table 2, Panel A presents summary statistics for the main sample. On average, $\Delta \ln SALES$ and $\Delta \ln XOPR$ are 0.1103 (median = 0.0796) and 0.1023 (median = 0.0781), respectively. The mean (median) values of our three strategy scores, $PRODUCT$, $CUSTOMER$, and $OPERATION$, are 0.0100 (−0.1521), 0.0180 (−0.2550), and 0.0198 (−0.2685), respectively. On average, our sample firms exhibit demand uncertainty of 0.3977 (median = 0.2460), asset intensity of 3.5960 (median = 1.1394), and employee intensity of 9.5188 (median = 4.6338). The average logarithm of firm ages in our sample is 2.6009 (median = 2.5649), and the average logarithm of market capitalization is 5.5238

(median = 5.5568). In general, the distribution of our sample characteristics is comparable to those reported in Banker et al. (2014), Anderson et al. (2003), and Irvine et al. (2016). Panel B provides correlations between our main variables. Almost all correlation coefficients for control variables are lower than 0.40, which is below the threshold of potential multi-collinearity (Gujarati, 2003).

4 Empirical Findings

4.1 Baseline Analysis

Table 3 reports the results for OLS regressions of cost structure on business strategies. Column (1) shows the results with industry and year fixed effects included before adding control variables. First, we find a significantly negative coefficient on $PRODUCT * \Delta \ln SALES$ (coefficient = −0.076, t -statistic = −10.59), suggesting that product leaders exhibit a more rigid cost structure. This is because their strong focus on innovation with fixed investments and severe concern about congestion risk will dominate their demand for flexibility in resource management and consideration of downside risk. Second, the coefficient on $CUSTOMER * \Delta \ln SALES$ (coefficient = 0.104, t -statistic = 15.70) is positive and significant, indicating a lower level of cost rigidity for customer-intimate firms. This result implies that it is more beneficial for them to use flexible and on-demand inputs to tailor offerings and limit downside risk. Third, we observe a positive and significant coefficient on $OPERATION * \Delta \ln SALES$ (coefficient = 0.125, t -statistic = 15.55). This result suggests that operational-excellence firms opt for flexible resource management, thus utilizing variable inputs to compete with rivals and relieve downside risk imposed by substitutes.

Column (2) shows the results after including additional control variables for cost structure. The estimated coefficient on $PRODUCT * \Delta \ln SALES$ remains negative and significant (coefficient = −0.061, t -statistic = −9.60). Meanwhile, we continue to find significantly positive coefficients on $CUSTOMER * \Delta \ln SALES$ (coefficient = 0.076, t -statistic = 13.07) and $OPERATION * \Delta \ln SALES$ (coefficient = 0.078, t -statistic = 12.19). The coefficients on most control variables show expected signs. For example, the coefficient on $UNCERT * \Delta \ln SALES$ is negative and significant (coefficient = −0.151, t -statistic = −13.66), suggesting that firms with greater demand uncertainty will increase committed capacity to relieve congestion at high demand realizations (Banker et al., 2014). The significantly negative coefficients on $AI * \Delta \ln SALES$ (coefficient = −0.002, t -statistic = −7.76) and $EI * \Delta \ln SALES$ (coefficient = −0.001, t -statistic = −3.17) imply that firms with higher adjustment costs of resources tend to exhibit a higher level of cost rigidity. Additionally, as in Irvine et al. (2016), we find that firm size is negatively associated with cost rigidity.

In columns (3) and (4), we include industry-year (Klasa et al., 2018) and firm fixed effects and find consistent results. Overall, our results in Table 3 suggest that, on average, firms

Table 3. The association between strategy and cost structure.

	Dependent variable = $\Delta \ln XOPR$							
	(1)		(2)		(3)		(4)	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
$\Delta \ln SALES$	0.516***	(69.66)	0.696***	(12.90)	0.702***	(12.96)	0.660***	(11.16)
PRODUCT*$\Delta \ln SALES$	-0.076***	(-10.59)	-0.061***	(-9.60)	-0.064***	(-9.98)	-0.063***	(-8.78)
CUSTOMER*$\Delta \ln SALES$	0.104***	(15.70)	0.076***	(13.07)	0.072***	(12.38)	0.080***	(11.55)
OPERATION*$\Delta \ln SALES$	0.125***	(15.55)	0.078***	(12.19)	0.078***	(12.12)	0.077***	(10.76)
UNCERT* $\Delta \ln SALES$			-0.151***	(-13.66)	-0.150***	(-13.56)	-0.165***	(-13.05)
AI* $\Delta \ln SALES$			-0.002***	(-7.76)	-0.002***	(-7.59)	-0.002***	(-7.21)
EI* $\Delta \ln SALES$			-0.001***	(-3.17)	-0.001***	(-3.34)	-0.000	(-0.60)
AGE* $\Delta \ln SALES$			-0.021***	(-2.60)	-0.020**	(-2.46)	-0.014	(-1.59)
MVE* $\Delta \ln SALES$			0.010***	(3.51)	0.009***	(3.14)	0.010***	(3.27)
GROWTH* $\Delta \ln SALES$			-0.039**	(-2.50)	-0.037**	(-2.38)	-0.036**	(-2.13)
LEV* $\Delta \ln SALES$			-0.024	(-1.12)	-0.019	(-0.87)	-0.030	(-1.24)
PRODUCT	0.011***	(6.97)	0.003**	(2.49)	0.003**	(2.56)	0.002	(0.74)
CUSTOMER	-0.008***	(-6.80)	-0.007***	(-6.70)	-0.007***	(-6.46)	-0.005***	(-2.71)
OPERATION	-0.010***	(-9.74)	-0.006***	(-6.96)	-0.006***	(-6.61)	-0.005***	(-3.44)
UNCERT			0.006	(1.41)	0.008*	(1.84)	0.000	(.)
AI			0.001***	(5.30)	0.001***	(5.15)	0.001***	(3.53)
EI			0.000	(1.51)	0.000	(1.52)	0.001***	(3.51)
AGE			-0.026***	(-15.80)	-0.025***	(-15.21)	-0.062***	(-10.86)
MVE			0.010***	(18.18)	0.010***	(17.84)	0.030***	(17.15)
GROWTH			0.066	(1.44)	0.060	(1.29)	0.063	(1.25)
LEV			-0.021***	(-3.88)	-0.022***	(-4.20)	-0.008	(-0.76)
Intercept	0.046***	(39.00)	0.037***	(6.94)	0.035***	(6.73)	0.016	(0.90)
Industry F.E.	Yes		Yes		No		No	
Year F.E.	Yes		Yes		No		Yes	
Industry x Year F.E.	No		No		Yes		No	
Firm F.E.	No		No		No		Yes	
N	62,116		62,116		62,116		60,779	
Adjusted R ²	0.4743		0.5531		0.5712		0.5612	

Notes: This table reports the results of estimating the association between strategy and cost structure. t-statistics are based on robust standard errors clustered at the firm level. Variable definitions are presented in Appendix A. All continuous variables are winsorized at the 1% and 99% levels. *, **, and *** denote significance based on two-tailed t-tests at or below the 10%, 5%, and 1% levels, respectively. AI = asset intensity; EI = employee intensity; MVE = market capitalization; LEV = leverage.

leaning toward a product leadership strategy exhibit a higher level of cost rigidity (*H1a*), while firms focusing on the customer intimacy and operational excellence strategies operate with a less rigid cost structure (*H2b* and *H3b*).

4.2 Validation Test: The Adoption of IDD

To address the concern that there is lack of variation in organizational strategy over time, we adopt a difference-in-difference framework by exploiting a trade secret law setting that provides a shift in strategy. Specifically, we use the staggered adoption of the IDD by U.S. state courts. The IDD is a legal doctrine that can grant the plaintiff firm an injunction to prevent a departing employee from working with a new employer in a similar position if the plaintiff demonstrates that the new employment would inevitably result in trade secrets revelation. Since IDD could enhance the legal

protection of trade secrets and promote the successful implementation of strategies through reducing employee mobility (Kim, 2014; Klasa et al., 2018; Li et al., 2018), we expect that IDD leads to a better alignment between cost structure and strategic orientation.¹⁵

We first verify that the adoption of IDD strengthens firms' orientation toward and implementation of existing business strategies. Following Klasa et al. (2018), we provide a list of precedent-setting cases adopting the IDD in Table A.4 (E-Companion) and use the dates of those cases to define the adoption year in a particular state. Table 4, Panel A shows that the mean and median values of the three strategy scores increase significantly from the pre-adoption period to the post-adoption period.

Next, we use a difference-in-difference method to examine how the adoption of IDD affects a firm's cost structure through the channel of business strategy. Following Klasa et al. (2018), we set an indicator variable, *IDD*, to be one after a firm's

Table 4. The association between strategy and cost structure: the IDD adoption.

Panel A: Strategy scores before and after the IDD adoption		Before IDD	After IDD	Diff	P-value
PRODUCT	Mean	−0.0131	−0.0163	−0.0032	.6850
	Median	−0.1905	−0.1752	0.0153*	.0826
CUSTOMER	Mean	−0.0445	0.0322	0.0767***	.0000
	Median	−0.3168	−0.2474	0.0694***	.0000
OPERATION	Mean	−0.0192	0.0029	0.0221***	.0008
	Median	−0.3201	−0.2918	0.0283***	.0092

Panel B: The association between strategy and cost structure after the IDD adoption				
Dependent variable = $\Delta \ln XOPR$				
	(1)		(2)	
	Staggered DiD regression		Stacked regression	
	Coef.	t-stat	Coef.	t-stat
$\Delta \ln SALES$	0.683***	(12.55)	0.666***	(30.66)
$PRODUCT * \Delta \ln SALES$	−0.048***	(−5.39)	−0.050***	(−18.71)
$CUSTOMER * \Delta \ln SALES$	0.067***	(7.62)	0.072***	(28.60)
$OPERATION * \Delta \ln SALES$	0.064***	(7.22)	0.076***	(27.24)
$IDD * \Delta \ln SALES$	0.024***	(2.02)	0.018***	(1.96)
$IDD * PRODUCT$	0.005***	(2.08)	0.004***	(2.53)
$IDD * CUSTOMER$	−0.001	(−0.69)	−0.003***	(−2.26)
$IDD * OPERATION$	−0.001	(−0.73)	0.001	(0.49)
$IDD * PRODUCT * \Delta \ln SALES$	−0.029**	(−2.42)	−0.024**	(−2.52)
$IDD * CUSTOMER * \Delta \ln SALES$	0.021*	(1.84)	0.021***	(2.65)
$IDD * OPERATION * \Delta \ln SALES$	0.027**	(2.24)	0.019*	(1.92)
$UNCERT * \Delta \ln SALES$	−0.151***	(−13.06)	−0.150***	(−32.27)
$AI * \Delta \ln SALES$	−0.003***	(−7.77)	−0.002***	(−13.06)
$EI * \Delta \ln SALES$	−0.000***	(−2.60)	−0.001***	(−11.13)
$AGE * \Delta \ln SALES$	−0.021***	(−2.47)	−0.031***	(−8.73)
$MVE * \Delta \ln SALES$	0.010***	(3.58)	0.007***	(5.87)
$GROWTH * \Delta \ln SALES$	−0.036***	(−2.31)	−0.049***	(−7.71)
$LEV * \Delta \ln SALES$	−0.023	(−1.07)	−0.045***	(−5.20)
PRODUCT	0.002	(1.00)	0.001**	(2.33)
CUSTOMER	−0.007***	(−4.15)	−0.005***	(−10.52)
OPERATION	−0.006***	(−4.16)	−0.008***	(−17.53)
IDD	−0.000	(−0.01)	0.005	(1.00)
UNCERT	0.006	(1.23)	0.003	(1.43)
AI	0.002***	(5.29)	0.001***	(9.17)
EI	0.000	(1.22)	0.000***	(6.19)
AGE	−0.026***	(−15.03)	−0.031***	(−39.11)
MVE	0.010***	(17.78)	0.014***	(54.47)
GROWTH	0.060	(1.30)	0.097***	(5.19)
LEV	−0.022***	(−3.92)	−0.032***	(−13.42)
Intercept	0.032	(0.66)	0.035***	(14.43)
Industry F.E.		Yes		Yes
Year F.E.		Yes		Yes
State F.E.		Yes		Yes
Stacked F.E.		No		Yes
N		60,364		333,357
Adjusted R ²		0.5563		0.5475

Continue.

headquarter state court adopts IDD, and zero otherwise.¹⁶ We revise model (1) by creating two-way and three-way interactions between *IDD*, $\Delta \ln SALES$, and strategy scores

and controlling for state-fixed effects as in Li et al. (2018). Panel B, column (1) shows that $IDD * PRODUCT * \Delta \ln SALES$ loads negatively (coefficient = −0.029, *t*-statistic = −2.42),

Table 4. Continued.

Panel C: Dynamics of the association between strategy and cost structure around the IDD adoption
Dependent variable = $\Delta \ln XOPR$

	Coef.	Coef.
$IDD_{-3} * PRODUCT * \Delta \ln SALES$	0.013	(0.25)
$IDD_{-2} * PRODUCT * \Delta \ln SALES$	0.081	(1.45)
$IDD_{-1} * PRODUCT * \Delta \ln SALES$	0.051	(0.67)
$IDD_0 * PRODUCT * \Delta \ln SALES$	-0.035	(-0.92)
$IDD_{+1} * PRODUCT * \Delta \ln SALES$	-0.069*	(-1.67)
$IDD_{+2} * PRODUCT * \Delta \ln SALES$	-0.027	(-0.63)
$IDD_{\geq+3} * PRODUCT * \Delta \ln SALES$	-0.024**	(-1.98)
$IDD_{-3} * CUSTOMER * \Delta \ln SALES$	-0.053	(-1.34)
$IDD_{-2} * CUSTOMER * \Delta \ln SALES$	0.016	(0.42)
$IDD_{-1} * CUSTOMER * \Delta \ln SALES$	0.183**	(2.14)
$IDD_0 * CUSTOMER * \Delta \ln SALES$	0.093***	(2.90)
$IDD_{+1} * CUSTOMER * \Delta \ln SALES$	-0.053*	(-1.69)
$IDD_{+2} * CUSTOMER * \Delta \ln SALES$	0.097***	(3.06)
$IDD_{\geq+3} * CUSTOMER * \Delta \ln SALES$	0.022*	(1.94)
$IDD_{-3} * OPERATION * \Delta \ln SALES$	-0.015	(-0.78)
$IDD_{-2} * OPERATION * \Delta \ln SALES$	-0.021	(-0.51)
$IDD_{-1} * OPERATION * \Delta \ln SALES$	0.023	(0.42)
$IDD_0 * OPERATION * \Delta \ln SALES$	-0.030	(-0.68)
$IDD_{+1} * OPERATION * \Delta \ln SALES$	-0.005	(-0.11)
$IDD_{+2} * OPERATION * \Delta \ln SALES$	-0.010	(-0.28)
$IDD_{\geq+3} * OPERATION * \Delta \ln SALES$	0.032**	(2.52)
Other controls		Yes
Industry F.E.		Yes
Year F.E.		Yes
State F.E.		Yes
N		60,364
Adjusted R ²		0.5572

Notes: This table reports strategy scores before and after the IDD adoption in Panel A, the results of estimating the association between strategy and cost structure following the IDD adoption in Panel B, and dynamics of changes in cost structure in Panel C. Industry, year, and state fixed effects are included in each model. *t*-statistics are based on robust standard errors clustered at the firm level. Variable definitions are presented in Appendix A. All continuous variables are winsorized at the 1% and 99% levels. *, **, and *** denote significance based on two-tailed *t*-tests at or below the 10%, 5%, and 1% levels, respectively. AI = asset intensity; EI = employee intensity; MVE = market capitalization; LEV = leverage; IDD = Inevitable Disclosure Doctrine.

while $IDD * CUSTOMER * \Delta \ln SALES$ (coefficient = 0.021, *t*-statistic = 1.84) and $IDD * OPERATION * \Delta \ln SALES$ (coefficient = 0.027, *t*-statistic = 2.24) load positively. This is consistent with our prediction that firms lean more toward their strategy after the IDD adoption and exhibit corresponding changes in cost structure. In addition, Baker et al. (2022) document that the staggered difference-in-difference test may be subject to estimation bias. Therefore, we implement a stacked regression to verify the robustness of the treatment effect estimates and report results in column (2).¹⁷ We find consistent results for the three strategies.

Finally, we conduct a parallel trends analysis to address the concern that our findings may be driven by concurrent changes or events other than the IDD adoption. We create indicator variables, IDD_{-3} , IDD_{-2} , and IDD_{-1} , for observations in 3 years, 2 years, and 1 year before the IDD adoption, respectively. Similarly, we create indicators, IDD_0 , IDD_{+1} ,

IDD_{+2} , and $IDD_{\geq+3}$ for observations in the current year, 1 year, 2 years, and 3 or more years after the IDD adoption. We include these indicators and corresponding interactions in model (1). Panel C shows that none of the coefficients on $IDD_{-3} * PRODUCT * \Delta \ln SALES$, $IDD_{-2} * PRODUCT * \Delta \ln SALES$, and $IDD_{-1} * PRODUCT * \Delta \ln SALES$ is significant, suggesting that IDD firms pursuing product leadership exhibit similar trends of cost structure as non-IDD firms before the IDD adoption (similar results for the other two strategies).¹⁸ These results also reveal the timing of changes in firms' cost structure after the adoption of IDD. We find that product leaders begin to increase cost rigidity 1 year after the IDD adoption, while this effect becomes stronger after 3 years. This result suggests that R&D investments and capacity additions are costly to adjust and take time to implement (Gaimon and Burgess, 2009). In contrast, firms respond more quickly to their enhanced orientation toward customer

intimacy driven by the IDD adoption: the coefficients on $IDD_0 * CUSTOM * \Delta \ln SALES$, $IDD_{+2} * CUSTOM * \Delta \ln SALES$, and $IDD_{\geq +3} * CUSTOM * \Delta \ln SALES$ are all positive and significant.¹⁹ Moreover, we find that the impact of IDD on cost structure through the operational excellence strategy starts to manifest in the third year after the IDD adoption, suggesting that it may take a few years to convert fixed resources into variable costs using real options or flexible operating models.²⁰

4.3 Validation Test: A Change Analysis

In this section, we conduct a change analysis to examine whether firms that made an internal strategic shift will exhibit corresponding changes in cost rigidity. We calculate changes in strategy scores over a 5-year window.²¹ For each strategy, we create a variable (*PRODUCT CHANGE*, *CUSTOMER CHANGE*, and *OPERATION CHANGE*) as the tercile rank of a firm's change in a strategy from year $t-5$ to t . Thus, we capture firms with a/an increased, unchanged, or decreased orientation toward a specific strategy. To measure changes in cost structure, we follow Irvine et al. (2016) and construct *ELAS IRVINE* as log changes in operating costs divided by log changes in sales. A higher value of *ELAS IRVINE* indicates a more elastic cost structure, while a lower value suggests a more rigid cost structure.²² We regress changes in cost structure from $t-5$ to t ($\Delta ELAS IRVINE$) on strategic shifts (*PRODUCT CHANGE*, *CUSTOMER CHANGE*, and *OPERATION CHANGE*) and changes in control variables over the same window.

Table 5 shows that the coefficient on *PRODUCT CHANGE* is negative and significant (coefficient = -0.048 , t -statistic = -2.19), indicating that firms leaning toward a product leadership strategy will exhibit an increase in cost rigidity. In addition, we find positive and significant coefficients on *CUSTOMER CHANGE* (coefficient = 0.043 , t -statistic = 2.04) and *OPERATION CHANGE* (coefficient = 0.040 , t -statistic = 1.79), suggesting that a shift toward customer intimacy or operational excellence will give rise to an increase in cost elasticity. Taken together, our results based on firms' internal strategic shifts provide supporting evidence on the impact of strategy on cost structure.²³

5 Additional Analyses

5.1 Strategy, Fixed Investments, and Demand Uncertainty

In Section 2.3, we conjecture that firms' competitive strategies influence the level of cost rigidity via strategic investments and demand conditions. To explore these channels, we first investigate whether strategy affects cost rigidity through fixed investments. Following Richardson (2006) and Biddle et al. (2009), we consider fixed investments in both capital and noncapital expenditures (R&D). Capital expenditures include funds used by a firm to acquire physical assets that

Table 5. The association between strategy and cost structure: a change analysis.

	Dependent variable = $\Delta ELAS IRVINE$	
	Coef.	t-stat
PRODUCT CHANGE	-0.048^{**}	(-2.19)
CUSTOMER CHANGE	0.043^{**}	(2.04)
OPERATION CHANGE	0.040^{*}	(1.79)
ΔAI	-0.007	(-1.32)
ΔEI	-0.000	(-0.00)
ΔAGE	-0.092	(-1.51)
ΔMVE	0.069^{***}	(4.27)
$\Delta GROWTH$	0.024	(1.35)
ΔLEV	0.016	(0.16)
Intercept	-0.054	(-0.68)
Year F.E.	Yes	
N	24,238	
Adjusted R^2	0.0015	

Notes: This table reports the results of estimating the association between strategy score increases and cost structure changes. Year-fixed effects are included. T-statistics are based on robust standard errors clustered at the firm level. Variable definitions are presented in Appendix A. All continuous variables are winsorized at the 1% and 99% levels. *, **, and *** denote significance based on two-tailed t-tests at or below the 10%, 5%, and 1% levels, respectively. AI = asset intensity; EI = employee intensity; MVE = market capitalization; LEV = leverage.

are considered permanent in nature. Meanwhile, R&D expenditures consist of investments in human and physical capital, which are costly to adjust in the short term and cause fixed costs. First, we calculate firms' fixed investments (*FIXED INV*) as the sum of capital and R&D expenditures deflated by lagged assets, and regress it on three strategy scores. Panel A of Table 6, column (1) shows a significantly positive coefficient on *PRODUCT* (coefficient = 0.024 , t -statistic = 11.07), suggesting that product leaders tend to undertake greater fixed investments. We also observe that the other two strategies are negatively associated with fixed investments (*CUSTOMER*: coefficient = -0.013 , t -statistic = -8.17 ; *OPERATION*: coefficient = -0.002 , t -statistic = -1.69). Next, in column (2), we investigate the association between three strategy scores and cost rigidity before controlling for *FIXED INV*. We continue to find that the product leadership strategy is associated with a higher level of cost rigidity, while the customer intimacy and operational excellence strategies are associated with a lower level of cost rigidity. Lastly, in column (3), we control for *FIXED INV* and its interaction with $\Delta \ln SALES$. We find that the coefficient on *FIXED INV* * $\Delta \ln SALES$ is significantly negative (coefficient: -0.107 , t -statistic = -8.61), indicating that firms' fixed inputs will lead to a high level of cost rigidity. We observe similar results on the association between strategy and cost structure. Altogether, our results in Panel A suggest that product leaders (customer-intimate and operational-excellence firms) undertake larger (smaller) fixed investments and hence exhibit a more (less) rigid cost structure.

Table 6. The association between strategy, fixed investments, and demand uncertainty.

Panel A: The Association between Strategy, Fixed Investments, and Cost Structure

	(1) FIXED INV		(2) $\Delta \ln XOPR$		(3) $\Delta \ln XOPR$	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
PRODUCT	0.024***	(11.07)	0.003**	(2.41)	−0.001	(−0.59)
CUSTOMER	−0.013***	(−8.17)	−0.007***	(−6.72)	−0.005***	(−4.66)
OPERATION	−0.002*	(−1.69)	−0.006***	(−6.95)	−0.006***	(−6.46)
UNCERT	0.068***	(10.57)	0.007	(1.45)	−0.006	(−1.40)
AI	−0.000**	(−2.27)	0.002***	(5.46)	0.001***	(5.22)
EI	0.001***	(11.33)	0.000	(1.30)	0.000	(0.14)
AGE	−0.046***	(−21.18)	−0.026***	(−15.83)	−0.021***	(−12.95)
MVE	0.004***	(5.44)	0.010***	(18.10)	0.010***	(17.31)
GROWTH	0.047***	(19.50)	0.066	(1.44)	0.029	(0.64)
LEV	−0.012	(−1.45)	−0.021***	(−3.89)	−0.022***	(−4.25)
$\Delta \ln SALES$			0.693***	(12.85)	0.733***	(13.80)
PRODUCT*$\Delta \ln SALES$			−0.061***	(−9.55)	−0.055***	(−8.80)
CUSTOMER*$\Delta \ln SALES$			0.077***	(13.09)	0.074***	(12.74)
OPERATION*$\Delta \ln SALES$			0.078***	(12.15)	0.075***	(12.20)
UNCERT* $\Delta \ln SALES$			−0.150***	(−13.56)	−0.141***	(−12.96)
AI* $\Delta \ln SALES$			−0.002***	(−7.66)	−0.002***	(−8.02)
EI* $\Delta \ln SALES$			−0.001***	(−3.22)	−0.000**	(−2.47)
AGE* $\Delta \ln SALES$			−0.021**	(−2.55)	−0.025***	(−3.15)
MVE* $\Delta \ln SALES$			0.010***	(3.53)	0.011***	(4.24)
GROWTH* $\Delta \ln SALES$			−0.039**	(−2.50)	−0.024	(−1.57)
LEV* $\Delta \ln SALES$			−0.024	(−1.10)	−0.013	(−0.61)
FIXED INV					0.156***	(16.23)
FIXED INV* $\Delta \ln SALES$					−0.107***	(−8.61)
Intercept	0.204***	(26.84)	0.036***	(6.91)	0.010*	(1.94)
Industry F.E.	Yes		Yes		Yes	
Year F.E.	Yes		Yes		Yes	
N	61,761		61,761		61,761	
Adjusted R ²	0.2496		0.5526		0.5617	

Panel B: The Association between Strategy, Demand Uncertainty, and Cost Structure

	(1) UNCERT		(2) $\Delta \ln XOPR$		(3) $\Delta \ln XOPR$	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
PRODUCT	0.016***	(3.95)	0.003**	(2.48)	0.003**	(2.49)
CUSTOMER	−0.021***	(−6.03)	−0.007***	(−6.58)	−0.007***	(−6.70)
OPERATION	−0.022***	(−8.06)	−0.007***	(−7.53)	−0.006***	(−6.96)
AI	0.007***	(14.68)	0.002***	(5.94)	0.001***	(5.30)
EI	0.002***	(6.72)	0.000**	(2.15)	0.000	(1.51)
AGE	−0.046***	(−11.03)	−0.030***	(−17.89)	−0.026***	(−15.80)
MVE	−0.045***	(−28.14)	0.011***	(19.46)	0.010***	(18.18)
GROWTH	0.107***	(41.54)	0.219***	(4.80)	0.066	(1.44)
LEV	−0.000	(−0.02)	−0.023***	(−4.19)	−0.021***	(−3.88)
$\Delta \ln SALES$			0.376***	(7.32)	0.696***	(12.90)
PRODUCT*$\Delta \ln SALES$			−0.062***	(−10.03)	−0.061***	(−9.60)
CUSTOMER*$\Delta \ln SALES$			0.082***	(14.09)	0.076***	(13.07)
OPERATION*$\Delta \ln SALES$			0.090***	(13.84)	0.078***	(12.19)
AI* $\Delta \ln SALES$			−0.003***	(−9.87)	−0.002***	(−7.76)
EI* $\Delta \ln SALES$			−0.001***	(−4.40)	−0.001***	(−3.17)
AGE* $\Delta \ln SALES$			−0.012	(−1.51)	−0.021***	(−2.60)
MVE* $\Delta \ln SALES$			0.018***	(6.46)	0.010***	(3.51)
GROWTH* $\Delta \ln SALES$			−0.102***	(−6.62)	−0.039**	(−2.50)
LEV* $\Delta \ln SALES$			−0.021	(−0.96)	−0.024	(−1.12)

Continue.

Table 6. Continued.

Panel B: The Association between Strategy, Demand Uncertainty, and Cost Structure						
	(1) <i>UNCERT</i>		(2) $\Delta \ln XOPR$		(3) $\Delta \ln XOPR$	
	Coef.	t-stat	Coef.	t-stat	Coef.	t-stat
<i>UNCERT</i>					0.006	(1.41)
<i>UNCERT</i> * $\Delta \ln SALES$					−0.151***	(−13.66)
Intercept	0.697***	(48.79)	0.046***	(9.48)	0.037***	(6.94)
Industry F.E.	Yes		Yes		Yes	
Year F.E.	Yes		Yes		Yes	
N	62,116		62,116		62,116	
Adjusted R ²	0.3727		0.5413		0.5531	

Notes: This table reports the results of estimating the association between strategy, fixed investments in physical capital and R&D, and cost structure in Panel A, and the association between strategy, demand uncertainty, and cost structure in Panel B. Industry and year fixed effects are included in each model. T-statistics are based on robust standard errors clustered at the firm level. Variable definitions are presented in Appendix A. All continuous variables are winsorized at the 1% and 99% levels. *, **, and *** denote significance based on two-tailed t-tests at or below the 10%, 5%, and 1% levels, respectively. AI = asset intensity; EI = employee intensity; MVE = market capitalization; LEV = leverage.

We also investigate whether demand uncertainty is a channel through which business strategy affects cost structure. We regress demand uncertainty (*UNCERT*) on strategy scores and report the results in Panel B of Table 6, column (1). The coefficient on *PRODUCT* is significantly positive (coefficient = 0.016, *t*-statistic = 3.95), while the coefficients on *CUSTOMER* and *OPERATION* are both significantly negative (*CUSTOMER*: coefficient = −0.021, *t*-statistic = −6.03; *OPERATION*: coefficient = −0.022, *t*-statistic = −8.06). These findings suggest that product leaders will realize more volatile demand associated with highly risky R&D projects. However, customer-intimate and operational-excellence firms' demand is more stable with reliable customer relationships or a narrow product focus. In column (2), before controlling for *UNCERT*, we find a positive (negative) association between the product leadership (customer intimacy and operational excellence) strategy and cost rigidity. In column (3), we include *UNCERT* and the corresponding interaction term as additional controls. We observe a negative and significant coefficient on *UNCERT** $\Delta \ln SALES$ (coefficient = −0.151, *t*-statistic = −13.66), consistent with Banker et al. (2014) that firms with more volatile demand will increase capacity to address congestion. We also obtain similar results on the association between strategy and cost rigidity. Overall, our results in Panel B confirm that product leaders (customer-intimate and operational-excellence firms) tend to experience more (less) volatile demand with higher (lower) congestion risk, thereby choosing a more (less) rigid cost structure.

5.2 Cross-Sectional Analyses

We also investigate the cross-sectional variations in the association between strategy and cost structure. First, we examine the effect of market shares. A firm's market share represents its proportion of total sales in the industry, where a lower market share indicates more intense competition because customers have more alternatives to choose from (Dhaliwal et al., 2016).

We find that the positive (negative) association between product leadership (the other two strategies) and cost rigidity is stronger when firms possess lower market share (Table A.5). That is, anticipating intense competition due to lower market shares, product leaders will undertake higher R&D investments to explore new opportunities, while customer-intimate and operational-excellence firms will decrease fixed investments to maintain flexibility.

Second, firms' demand conditions may affect their choices of capacity levels and cost structure. Demand uncertainty, as captured by the standard deviation of sales changes, indicates a higher likelihood of both usually high and unusually low demand (Banker et al., 2014). We observe that the positive (negative) association between product leadership (the other two strategies) and cost rigidity is stronger for firms with higher demand uncertainty (Table A.5). This is because, with volatile demand, product leaders are more concerned about the upside congestion risk, while firms following the other two strategies emphasize more on the downside default risk. Overall, the evidence suggests that firms with lower market shares and higher demand uncertainty are more likely to align cost structures with strategies.

5.3 The Effect of Size

As discussed in Section 2.3, operational-excellence firms could take advantage of economies of scales by increasing cost rigidity or achieve resource flexibility by decreasing cost rigidity. Our main findings suggest that, on average, a firm's orientation toward operational excellence is associated with a less rigid cost structure. However, firm size may play a significant role in determining this association. As economies of scale can be achieved by spreading fixed costs over a higher volume of output, it would be more feasible for larger firms to reach the scale necessary to lower unit costs. Therefore, operational-excellence firms above a certain size may rely on more rigid costs to pursue cost reduction and minimization. To explore

this possibility, we limit our sample to firms in the top 5% size (*MVE*) group and investigate the association between the operational excellence strategy and cost rigidity. We find a negative and significant coefficient on *OPERATION*ΔlnSALES* (Table A.6). This result suggests that firms with relatively large size could achieve operational excellence through substantial fixed investments and mass production.

6 Conclusions

In this paper, we use a direct measure of strategy based on textual analysis of public disclosures and investigate its impact on cost structure. Building on the strategy typology in Miller (1986) and Treacy and Wiersema (1993), we find that firms leaning toward product leadership choose a more rigid cost structure, while firms pursuing customer intimacy and operational excellence select a more elastic cost structure. Our results are robust to a difference-in-difference research design using the adoption of IDD as a quasirational shock to corporate

strategies and a change analysis based on internal strategic shifts. We also conduct a set of additional tests and show that strategy impacts cost structure through the channels of strategic investments and demand conditions.

Overall, this study highlights business strategy as a new determinant of firms' cost structure decisions, which have significant implications for financial performance and operations management. Our findings further contribute to the literature by documenting the real effects of strategy on corporate resource allocation and capacity management policies in multiple functional departments, thereby improving our understanding of the execution of strategy and coordination between departments. This study suffers from a potential limitation. Although the disclosure of business strategy is mandated by Regulation S-K, the exact information content of this disclosure is subject to managerial discretion. Future research may seek to distinguish the discretionary and nondiscretionary information contents conveyed by managers and investigate whether they impact cost structure differently.

Appendix A. Variable definitions

$\Delta \ln \text{SALE}$	Log-changes in sales revenue of firm <i>i</i> from year <i>t</i> -1 to year <i>t</i> .
$\Delta \ln \text{XOPR}$	Log-changes in operating costs of firm <i>i</i> from year <i>t</i> -1 to year <i>t</i> .
PRODUCT	Strategy score for product leadership calculated from factor analysis based on counts of keywords scaled by total number of words in Item 1-Business of 10-K filing for firm <i>i</i> in year <i>t</i> .
CUSTOMER	Strategy score for customer intimacy calculated from factor analysis based on counts of keywords scaled by total number of words in Item 1-Business of 10-K filing for firm <i>i</i> in year <i>t</i> .
OPERATION	Strategy score for operational excellence calculated from factor analysis based on counts of keywords scaled by total number of words in Item 1-Business of 10-K filing for firm <i>i</i> in year <i>t</i> .
UNCERT	The standard deviation of $\Delta \ln \text{SALE}$ for all valid observations of firm <i>i</i> .
AI	Total assets scaled by total sales for firm <i>i</i> in year <i>t</i> .
EI	The number of employees scaled by total sales for firm <i>i</i> in year <i>t</i> .
AGE	Natural log of firm age, measured from the first year firm <i>i</i> appears in the Compustat Annual Fundamentals database.
MVE	Natural log of market capitalization for firm <i>i</i> in year <i>t</i> .
GROWTH	Annual sales growth for firm <i>i</i> in year <i>t</i> .
LEV	Long-term liabilities scaled by total assets for firm <i>i</i> in year <i>t</i> .
IDD	An indicator variable equal to one after a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₋₃	An indicator variable equal to one in three years before a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₋₂	An indicator variable equal to one in two years before a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₋₁	An indicator variable equal to one in one year before a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₀	An indicator variable equal to one in the current year that a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₊₁	An indicator variable equal to one in one year after a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD ₊₂	An indicator variable equal to one in two years after a firm's headquarter state court recognizes IDD, and zero otherwise.
IDD _{≥+3}	An indicator variable equal to one in three or more years after a firm's headquarter state court recognizes IDD, and zero otherwise.
PRODUCT CHANGE	The tercile rank of a firm's change in PRODUCT from year <i>t</i> -5 to <i>t</i> .
CUSTOMER CHANGE	The tercile rank of a firm's change in CUSTOMER from year <i>t</i> -5 to <i>t</i> .
OPERATION CHANGE	The tercile rank of a firm's change in OPERATION from year <i>t</i> -5 to <i>t</i> .
ELAS IRVINE	Log-changes in operating costs divided by log-changes in sales from year <i>t</i> -1 to <i>t</i> .
$\Delta \text{ELAS IRVINE}$	Changes in ELAS IRVINE from year <i>t</i> -5 to <i>t</i> .
ΔAI	Changes in AI from year <i>t</i> -5 to <i>t</i> .
ΔEI	Changes in EI from year <i>t</i> -5 to <i>t</i> .
ΔAGE	Changes in AGE from year <i>t</i> -5 to <i>t</i> .
ΔMVE	Changes in MVE from year <i>t</i> -5 to <i>t</i> .
ΔGROWTH	Changes in GROWTH from year <i>t</i> -5 to <i>t</i> .
ΔLEV	Changes in LEV from year <i>t</i> -5 to <i>t</i> .
FIX INV	The sum of capital and R&D expenditures scaled by lagged total assets for firm <i>i</i> in year <i>t</i> .

Appendix B. Keyword list (* denotes wildcards)

I. Differentiation strategy	II. Product	III. Market	IV. Customer
1) <i>different*</i> (Campbell-Hunt, 2000; Porter, 1985; Thornhill and White, 2007)	1) <i>innovate*</i> (Campbell-Hunt, 2000; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Shinkle et al., 2013; Thornhill and White, 2007; White, 1986; Zatzick et al., 2012)	1) <i>marketing*</i> (Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993; Porter, 1985; White, 1986; Zahra and Covin, 1993)	1) <i>customer* service*</i> (Hambrick, 1983a; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993; Porter, 1985)
2) <i>unique*</i> (Dess and Davis, 1984; Hambrick, 1983a; Kotha and Vadlamani, 1995; Porter, 1985; White, 1986; Zahra and Covin, 1993)	2) <i>creativ*</i> (White, 1986)	2) <i>advertis*</i> (Nayyar, 1993; Kotha and Vadlamani, 1995; Zahra and Covin, 1993)	2) <i>consumer* service*</i> (Nayyar, 1993)
3) <i>superior*</i> (Campbell-Hunt, 2000)	3) <i>research and development, R&D</i>	3) <i>brand*</i> (Hambrick, 1983a; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993; Porter, 1985; Zahra and Covin, 1993)	3) <i>customer* need*</i> (White, 1986)
4) <i>premium*</i> (Nayyar, 1993; Porter, 1985)	4) <i>techni*</i> (White, 1986; Thornhill and White, 2007; Zatzick et al., 2012)	4) <i>reputation*</i> (Nayyar, 1993)	4) <i>sales support*</i> (White, 1986)
5) <i>high* pric*</i> (Dess and Davis, 1984; Nayyar, 1993; White, 1986; Zahra and Covin, 1993)	5) <i>technology*</i>	5) <i>trademark*</i>	5) <i>post-purchase service*</i> (White, 1986)
6) <i>high* margin</i> (Porter, 1985)	6) <i>patent*</i> (Miller and Friesen, 1986a, 1986b)		6) <i>customer preference*</i> (Campbell-Hunt, 2000; Zatzick et al., 2012)
7) <i>inelasticity</i> (Miller and Friesen, 1986a; Miller and Friesen, 1986b)	7) <i>proprietary*</i>		7) <i>consumer* preference*</i> (Campbell-Hunt, 2000; Zatzick et al., 2012)
8) <i>excellen*</i>	8) <i>new* product*</i> (Nayyar, 1993; Porter, 1985; Zatzick et al., 2012)		8) <i>consumer* relation*</i> , <i>consumer* experience*</i> , <i>consumer* support*</i>
9) <i>leading edge</i>			<i>loyalty*</i> (Miller and Friesen, 1986a; Miller and Friesen, 1986b)
10) <i>upscale</i>			9) <i>customiz*</i>
			10) <i>tailor*</i> (White, 1986)
			11) <i>personaliz*</i>

(continue)

Appendix B. Continued.

V. Human resource (Porter, 1985)		VI. Cost strategy		VII. Cost Control		VIII. Operational efficiency	
1) <i>talent*</i>	1) <i>cost leader*</i> (Porter, 1985)	1) <i>control*</i> <i>cost*</i> , <i>control*</i>	1) <i>control*</i> <i>cost*</i> , <i>control*</i>	1) <i>control*</i> <i>cost*</i> , <i>control*</i>	1) <i>control*</i> <i>cost*</i> , <i>control*</i>	1) <i>efficient*</i> (Miller and Friesen, 1986a, Miller and Friesen, 1986b; Nayyar, 1993; Kotha and Vadlamani, 1995; Shinkle et al., 2013; White, 1986; Zahra and Covin, 1993; Zatzick et al., 2012)	1) <i>efficient*</i> (Miller and Friesen, 1986a, Miller and Friesen, 1986b; Nayyar, 1993; Kotha and Vadlamani, 1995; Shinkle et al., 2013; White, 1986; Zahra and Covin, 1993; Zatzick et al., 2012)
2) <i>train*</i>	2) <i>low* price*</i>	2) <i>minimize*</i> <i>cost*</i> , <i>minimize*</i>	2) <i>minimize*</i> <i>cost*</i> , <i>minimize*</i>	2) <i>minimize*</i> <i>cost*</i> , <i>minimize*</i>	2) <i>minimize*</i> <i>cost*</i> , <i>minimize*</i>	2) <i>high* yield*</i> (White, 1986)	2) <i>high* yield*</i> (White, 1986)
3) <i>skill*</i>	3) <i>low* cost*</i> (Dess and Davis, 1984; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Porter, 1985; White, 1986)	3) <i>reduce* cost*</i> , <i>reduce* expense*</i> , <i>reduce* overhead*</i> (Campbell-Hunt, 2000; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993)	3) <i>reduce* cost*</i> , <i>reduce* expense*</i> , <i>reduce* overhead*</i> (Campbell-Hunt, 2000; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993)	3) <i>reduce* cost*</i> , <i>reduce* expense*</i> , <i>reduce* overhead*</i> (Campbell-Hunt, 2000; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993)	3) <i>reduce* cost*</i> , <i>reduce* expense*</i> , <i>reduce* overhead*</i> (Campbell-Hunt, 2000; Miller and Friesen, 1986a; Miller and Friesen, 1986b; Nayyar, 1993)	3) <i>process* improvement*</i> (Nayyar, 1993)	3) <i>process* improvement*</i> (Nayyar, 1993)
4) <i>intellectual* property*</i>	4) <i>cost advantage*</i> (Porter, 1985)	4) <i>monitor* cost*</i> , <i>monitor* expense*</i> , <i>monitor* overhead*</i>	4) <i>monitor* cost*</i> , <i>monitor* expense*</i> , <i>monitor* overhead*</i>	4) <i>monitor* cost*</i> , <i>monitor* expense*</i> , <i>monitor* overhead*</i>	4) <i>monitor* cost*</i> , <i>monitor* expense*</i> , <i>monitor* overhead*</i>	4) <i>asset* utilization*</i>	4) <i>asset* utilization*</i>
5) <i>human capital*</i>	5) <i>competitive price*</i> (White, 1986; Zahra and Covin, 1993)	5) <i>save* cost*</i> , <i>save* expense*</i> , <i>save* overhead*</i>	5) <i>save* cost*</i> , <i>save* expense*</i> , <i>save* overhead*</i>	5) <i>save* cost*</i> , <i>save* expense*</i> , <i>save* overhead*</i>	5) <i>save* cost*</i> , <i>save* expense*</i> , <i>save* overhead*</i>	5) <i>capacity* utilization*</i> (Zahra and Covin, 1993)	5) <i>capacity* utilization*</i> (Zahra and Covin, 1993)
	6) <i>aggressive price*</i> (Miller and Friesen, 1986a; Miller and Friesen, 1986b)	6) <i>improve* cost*</i>	6) <i>improve* cost*</i>	6) <i>improve* cost*</i>	6) <i>improve* cost*</i>		
		7) <i>cost*</i> <i>control*</i> , <i>cost*</i> <i>improvement*</i> , <i>cost*</i> <i>minimization*</i> , <i>cost*</i> <i>reduction*</i> , <i>cost*</i> <i>saving*</i> (Dess and Davis, 1984; Hambrick, 1983a; Kotha and Vadlamani, 1995; Thornhill and White, 2007)	7) <i>cost*</i> <i>control*</i> , <i>cost*</i> <i>improvement*</i> , <i>cost*</i> <i>minimization*</i> , <i>cost*</i> <i>reduction*</i> , <i>cost*</i> <i>saving*</i> (Dess and Davis, 1984; Hambrick, 1983a; Kotha and Vadlamani, 1995; Thornhill and White, 2007)	7) <i>cost*</i> <i>control*</i> , <i>cost*</i> <i>improvement*</i> , <i>cost*</i> <i>minimization*</i> , <i>cost*</i> <i>reduction*</i> , <i>cost*</i> <i>saving*</i> (Dess and Davis, 1984; Hambrick, 1983a; Kotha and Vadlamani, 1995; Thornhill and White, 2007)	7) <i>cost*</i> <i>control*</i> , <i>cost*</i> <i>improvement*</i> , <i>cost*</i> <i>minimization*</i> , <i>cost*</i> <i>reduction*</i> , <i>cost*</i> <i>saving*</i> (Dess and Davis, 1984; Hambrick, 1983a; Kotha and Vadlamani, 1995; Thornhill and White, 2007)		
		8) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	8) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	8) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	8) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>		
		9) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	9) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	9) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	9) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>		
		10) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	10) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	10) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>	10) <i>expense* control*</i> , <i>expense* improvement*</i> , <i>expense* minimization*</i> , <i>expense* reduction*</i> , <i>expense* saving*</i>		
		11) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	11) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	11) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>	11) <i>overhead* control*</i> , <i>overhead* improvement*</i> , <i>overhead* minimization*</i> , <i>overhead* reduction*</i> , <i>overhead* saving*</i>		

Appendix C. Factor analysis of strategy measures

Keywords	Estimated factor loadings			Communality
	Factor 1	Factor 2	Factor 3	
<i>Differentiation strategy</i>	0.2398	0.6017	0.0804	0.4260
<i>Product</i>	0.0747	0.0108	0.7398	0.5531
<i>Market</i>	−0.1747	0.7875	−0.0087	0.6508
<i>Customer</i>	0.2884	0.6077	0.1649	0.4797
<i>Human resources</i>	0.0097	0.0869	0.7655	0.5937
<i>Cost strategy</i>	0.6507	0.0923	−0.0846	0.4391
<i>Cost control</i>	0.6994	−0.0495	0.036	0.4930
<i>Operational efficiency</i>	0.6473	0.0326	0.2035	0.4615
Eigenvalue	1.5086	1.3712	1.2170	
Cronbach's alpha score	0.5011	0.4878	0.4525	
Label of extracted factors	OPERATION	CUSTOMER	PRODUCT	

Notes: This table reports the results of an exploratory factor analysis with varimax rotation based on textual data of 71,981 observations from 1993 to 2019. We retain three output factors with eigenvalues greater than one. Bold figures indicate keywords with factor loadings greater than 0.60. The keyword selection procedure is presented in Appendix B.

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Notes

1. The management literature provides various typologies of organizational strategy. In this paper, we adopt the typology proposed by Miller (1986) and Treacy and Wiersema (1995) based on the results of our factor analysis with minimum human intervention.
2. Costs are driven by activity resources and cost behavior reflects resource adjustment in response to activity changes. Since we

cannot observe underlying resource commitments directly, we infer the proportion of fixed and variable costs in the cost structure from the short-run cost response to changes in dollar sales. Previous research in this area commonly employs similar specifications (Aboody et al., 2018; Anderson et al., 2003; Banker et al., 2014; Kallapur and Eldengburg 2005; Noreen and Soderstrom, 1994, 1997).

3. Existing proxies of strategy are often constructed based on executive surveys (Slater and Olson, 2001), proprietary databases (Hambrick, 1983a; Hambrick, 1983b), and expert panelists' evaluation (Dess and Davis, 1984; Zott and Amit, 2008), which are subject to human intervention and constrained to a small sample of companies. At the same time, some accounting studies use easily obtainable accounting variables, such as sales and cost measures, to quantify business strategies (Ittner et al., 1997; Bentley et al., 2013). However, this type of measures only captures strategy indirectly and imprecisely because accounting numbers can be influenced by other factors such as macroeconomic changes and regulatory shifts.
4. Hall and Lerner (2010) document that salaries paid to skilled and well-educated scientists and engineers account for more than half of R&D spending in practice.
5. For example, in early years, Apple conveyed its commitment to innovation by investing in fixed costs consistently, which encouraged customers to purchase its computers and software companies to develop operating system compatible software.
6. See a press release by Hewlett-Packard in 2002 (<https://www.forbes.com/sites/panosmourdoukoutas/2014/05/23/whats-haunting-hewlett-packard-outsourcing-bad-acquisitions-and-lack-of-innovation/?sh=2a1063c3608c>). Similarly, Sony outsourced its LCD TV production to Hon Hai and concentrated internal resources on R&D to achieve product differentiation. Through leveraging external manufacturing resources, Sony could reduce fixed costs and improve profitability (<https://www.computerworld.com/article/2765989/sony-to-outsource-lcd-tv-production-to-hon-hai.html>).
7. We start from 1993 because all U.S. public companies were phased into EDGAR filing over a three-year period ending on May 6, 1996. Since then, all 10-K filings are available to the

- public and can be downloaded online from the SEC's EDGAR website.
8. Our results are robust if we include Item 1 sections that are shorter than 1,000 words.
 9. We use wildcards (indicated by asterisk) to capture keyword variations. For two-word phrases, we count the form both with and without a hyphen between the two words. Additionally, for phrases that include a verb and its object in the same sentence, we do not require them to be adjacent to each other.
 10. The management literature on organizational strategy documents that strategy typologies, such as differentiation and cost leadership, are conceptually and theoretically independent from each other because they establish different focus and future directions for organizations (Porter, 1980; Treacy and Wiersema, 1993). Accordingly, the word libraries based on the management literature for each distinct strategy are also independent from each other. The differences in the eight keyword groups will result in low correlations among the output factors. Therefore, we use an orthogonal rotation method (varimax) in the factor analysis. Consistent with this notion, Panel B of Table 2 shows that the correlations among the three strategy scores are all below 0.08. Our main results remain unchanged if we adopt an oblique rotation method.
 11. We label the first factor as *OPERATION* as three groups of keywords, i.e., Cost Control, Cost Strategy, and Operational Efficiency exhibit high loadings (greater than 0.60). Next, we label the second factor as *CUSTOMER* because keywords in Market, Differentiation Strategy, and Customer exhibit loadings greater than 0.60. Lastly, the third factor is labeled as *PRODUCT*, on which Human Resource and Product load high (greater than 0.60). The Cronbach alpha scores for the three output factors are 0.5011, 0.4878, and 0.4525, respectively. We acknowledge that they are lower than 0.60, i.e., the minimum threshold for standard survey-based scale measurements (DeVellis, 2012). Therefore, to validate the behavior of our constructs, we use them to replicate prior findings on strategic fit along supply chain as in Chang et al. (2022). Appendix A.1 in the E-Companion presents a detailed description of the replication test. We obtain results consistent with Chang et al. (2022), thus providing some credence to the reliability of our constructs.
 12. Weighting factor scores based on factor loadings instead of weighting equally mitigate the concern of repeatedly counting highly correlated words.
 13. Although strategy typologies are conceptually independent from each other, some companies may pursue multiple strategies to varying degrees (Treacy and Wiersema, 1993; Bentley et al., 2013). Our strategy measure can capture a firm's orientation toward the three strategies instead of classifying its strategy exclusively into one. Table A.1 in the E-Companion shows that 9,352 observations (15.06% of the full sample) exhibit high scores in all three strategies, while some firms (11.73%, 11.77%, and 11.59%) score high in two strategies. In addition, we identify S&P 500 firms that are ranked the highest for each of the three strategies. Table A.2 in the E-companion presents this list and shows that it is consistent with the public recognition of these firms' strategic positioning.
 14. We validate our multidimensional strategy scores by checking the correlation between our strategy scores with the strategy measure proposed by Bentley et al. (2013). The measure in Bentley et al. (2013) is constructed using accounting variables (hereafter *STRATEGY*). A higher value of it indicates prospectors (similar to product leaders), while a lower value suggests defenders (similar to operational-excellence firms). We find a positive and significant correlation between *STRATEGY* and *PRODUCT* (0.2087), but the correlation between *STRATEGY* and *OPERATION* is much smaller (0.0103).
 15. For example, product leaders rely heavily on high-skilled employees with proprietary knowledge to maintain innovation (Bentley et al., 2013; Miles and Snow, 2003). As the IDD legislation forces the retention of key talents and reduces the risk of losing proprietary knowledge, product leaders are more willing to pursue innovation with continuous investments in human capital. Similarly, customer-intimate and operational-excellence firms are less concerned about the leakage of confidential information possessed by key employees (such as customer lists for customer-intimate firms and cost-efficient technology information for operational-excellence firms) after IDD.
 16. There are three state courts that adopt the IDD at first and reject it later (Florida in 2001, Michigan in 2002, and Texas in 2003). For firms headquartered in these three states, we set the IDD indicator to zero in years after the rejection.
 17. Following Baker et al. (2022), we create event-specific "clean 2x2" datasets. For each dataset, we include all observations from a cohort that adopts IDD in the same year as "treated" firms and never-treated observations as "clean" controls. Then we append these datasets together and estimate a stacked DiD regression with dataset-specific and time fixed effects. The essence of this method is to estimate the DiD for each dataset and then combine the treatment effects across cohorts using variance weighting.
 18. One exception is that the coefficient on $IDD_{-1} * CUSTOMER * \Delta \ln SALES$ is positive and significant. To investigate whether the overall effect of customer intimacy on cost structure before the IDD adoption is significant, we conduct an F-test for $IDD_{-3} * CUSTOMER * \Delta \ln SALES + IDD_{-2} * CUSTOMER * \Delta \ln SALES + IDD_{-1} * CUSTOMER * \Delta \ln SALES$. We find that the sum of the coefficients is not significantly different from zero with a p -value of 0.2069.
 19. We notice that in one year after the IDD adoption, the coefficient on $IDD_{+1} * CUSTOMER * \Delta \ln SALES$ becomes negative (coefficient = -0.053, t -statistic = -1.69). This result is consistent with our argument that strategic investments aligned with customer intimacy could be adjusted flexibly. An F-test for $IDD_0 * CUSTOMER * \Delta \ln SALES + IDD_{+1} * CUSTOMER * \Delta \ln SALES + IDD_{+2} * CUSTOMER * \Delta \ln SALES + IDD_{\geq +3} * CUSTOMER * \Delta \ln SALES$ suggests that the sum of the coefficients is different from zero with a p -value of .0096, confirming that the overall effect aligns with our main findings.
 20. In Appendix A.2 in the E-Companion, Figures A.1, A.2, and A.3 plot the estimates of treatment effects from $t-3$ to $t+3$ around the IDD adoption for *PRODUCT*, *CUSTOMER*, and *OPERATION*, respectively. For the product leadership strategy, Figure A.1 shows a downward trend after the adoption of IDD. Meanwhile, for the customer intimacy and operational excellence strategies, we observe positive effects of the IDD adoption. Specifically, the treatment effects become positive since year t for customer intimacy (Figure A.2) and since year $t+3$ for operational excellence (Figure A.3).
 21. Business strategy is a comprehensive plan that defines goals for the long run. Once firms employ a particular strategy, they should

design appropriate organizational structure and commit supporting resources over a sufficiently long-time horizon (Miles and Snow, 1978, 2003). Therefore, we expect that strategy scores will remain relatively stable over time. We indeed find that the three strategy scores are positively and significantly correlated with their own lagged values. Specifically, the correlation coefficients are all above 0.80 and significant at one percent level (*PRODUCT*: 0.9030; *CUSTOMER*: 0.8918; *OPERATION*: 0.8277), supporting the persistence of strategy over time. As the annual changes in strategy scores may be small and insignificant, we calculate changes in strategy scores over a five-year window.

22. In Appendix A.7 in the E-Companion, we use *ELAS IRVINE* in a robustness test and find consistent results on the association between strategy and cost rigidity.
23. Note that we exclude $\Delta UNCERT$ in this change regression because *UNCERT* is a firm-level measure calculated as the standard deviation of sales changes for all valid observations of a firm (Banker et al., 2014). In a sensitivity test, we also define demand uncertainty as the standard deviation of sales changes for prior three years or five years. We obtain similar results, albeit with weaker significance. This may be because our sample is reduced significantly by requiring the 5-year measurement window.

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